



Grade 8 SS Geography

2. Elements of Weather

Learning Guide

Differences Between Weather and Climate : Recap

Weather	Climate
Short-term	Long-term — usually 30 years or more
Changes daily or hourly	Changes over many years
Example: Today is rainy, tomorrow is sunny	Example: Durban has a warm and wet climate
Can be forecast	Hard to predict daily, but shows trends

Main Elements of Weather

The four main **elements of weather** are:

1. **Temperature**
2. **Humidity**
3. **Wind**
4. **Precipitation**

1. Temperature

- **Temperature** measures how **hot or cold the air is**.
- It is measured using a **thermometer**.
- Temperature influences what people:
 - **Wear**
 - **Eat**
 - **Do daily**

2. Humidity

- **Humidity** is the amount of **water vapour** or **moisture** in the air.
- It is measured using a **hygrometer**.

High Humidity

- Makes people feel **hotter**.
- Causes sweat not to **evaporate easily**.

Low Humidity

- Makes the air feel **dry**.

3. Wind

- **Wind** is the movement of air from **high-pressure areas** to **low-pressure areas**.
- Wind is measured using:
 - **Anemometer** — measures wind **speed**
 - **Wind vane** — measures wind **direction**

Importance of Wind

Wind affects:

- **Temperature**
- **Weather patterns**
- **Plant growth**

4. Precipitation

- **Precipitation** is any form of **water falling from the sky**.
 - It includes:
 - **Rain**
 - **Snow**
 - **Hail**
 - **Sleet**
 - It is measured using a **rain gauge**.
 - Precipitation is vital for:
 - **Agriculture**
 - **Water supply**
-

Key Terms

Term	Meaning
Weather	Day-to-day atmospheric conditions
Climate	Average weather over a long period
Temperature	How hot or cold the air is
Humidity	Amount of water vapour in the air
Wind	Movement of air from high pressure to low pressure
Precipitation	Water falling from the sky
Thermometer	Instrument used to measure temperature
Hygrometer	Instrument used to measure humidity
Anemometer	Instrument used to measure wind speed
Wind vane	Instrument used to measure wind direction
Rain gauge	Instrument used to measure precipitation

For the full interactive presentation, visit May Learning Hub:
www.maylearninghub.co.za